

Understanding & Boost Consumer Behavior for Online Shopping

Hoai Thuong Tran

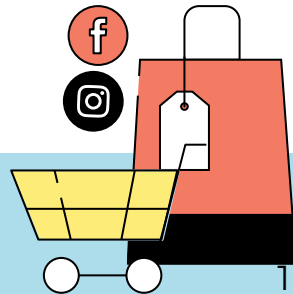
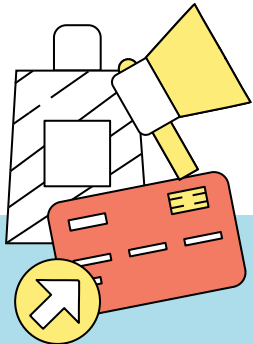


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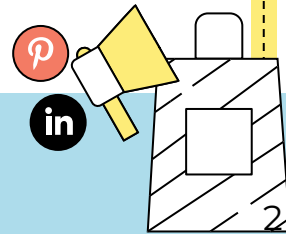
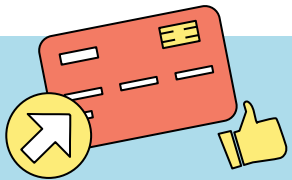
Methodology

03

**Analysis &
Insights**

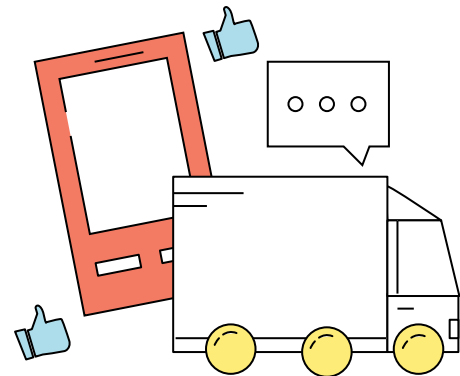
04

Conclusion

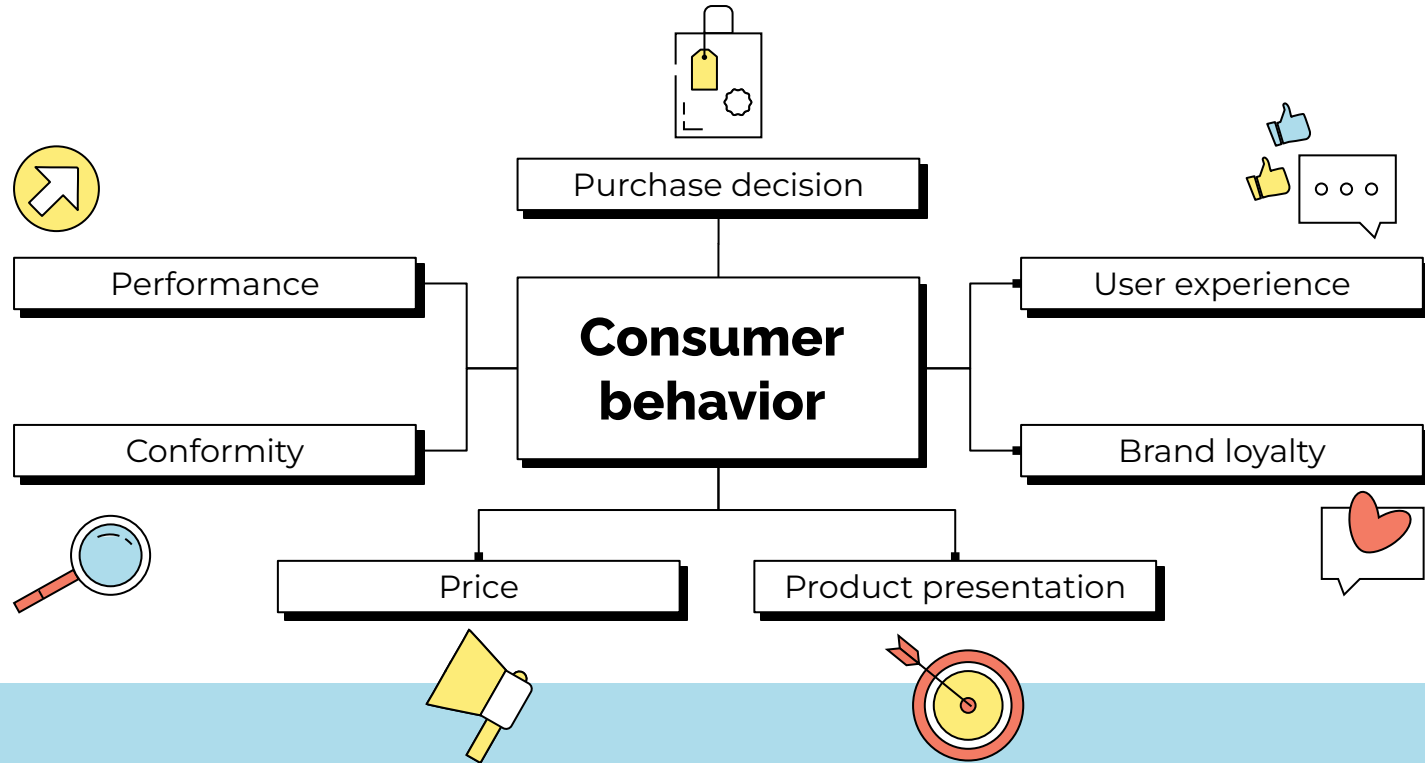


01

Introduction



Introduction



Online shopping is growing very fast. So, it is very important to understand how customers behave online

Objectives

Understanding

Understand customer behavior and engagement patterns by analyzing how customers browse and interact to identify what keeps them engaged.

Prediction

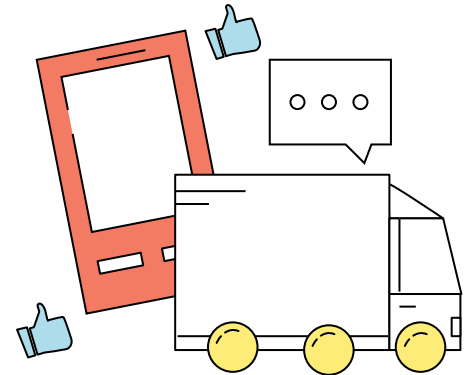
Use machine learning to predict and improve user engagement, improve the shopping experience and marketing effectiveness.

Clickstream Data for Online Shopping Dataset records clickstream activity from an online store selling clothing for pregnant women over five months in 2008.

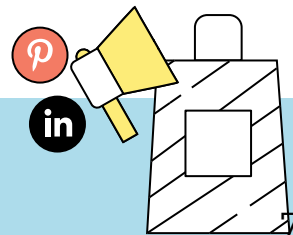
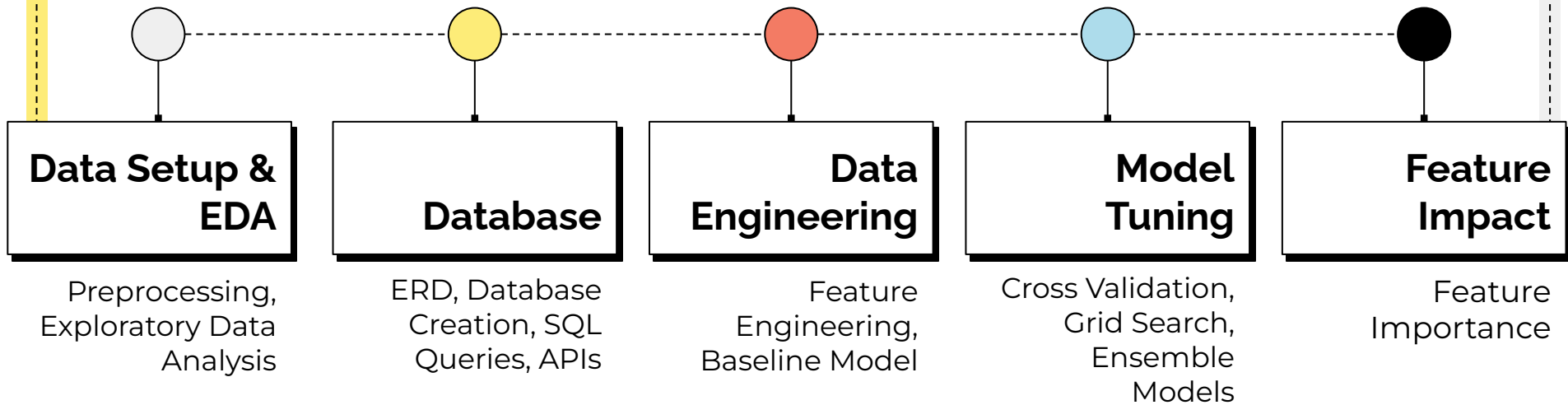


02

Methodology



Methodology



Planning

Final Project Hoai Thuong 000 ▾

Project Resources → ← ...

✓ Datasets Selection

+ Add a card

To Do → ← ...

Report RNCP 25 pages
🕒 Feb 1 - Feb 11

Slides RNCP
🕒 Feb 1 - Feb 12

+ Add a card

Pending → ← ...

Tableau Visualizations
🕒 Feb 7

Database Creation, ERD
🕒 Feb 9

SQL Queries
🕒 Feb 9

Exposing Data via APIs
🕒 Feb 10

+ Add a card

Done → ← ...

✓ Dataset Understanding
🕒 Jan 21 📧 1/1

✓ Data Cleaning
🕒 Jan 21 📧 3/3

✓ Exploratory Data Analysis (EDA)
🕒 Jan 25 📧 5/5

✓ Feature Engineering
🕒 Jan 26 📧 2/2

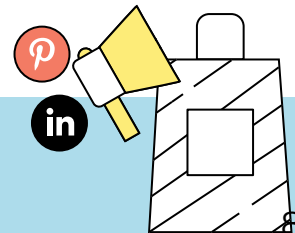
✓ Evaluation Baseline Model (KNN)
🕒 Jan 27

✓ Hyperparameter Tuning Baseline Model
🕒 Jan 28

✓ Evaluation (Fine-Tuned) Models & Comparison with Baseline Model
🕒 Jan 28 📧 2/2

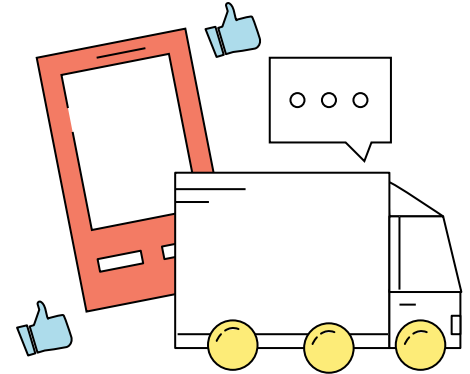
✓ Feature Impact (Gradient Boosting)
🕒 Jan 29

+ Add a card



03

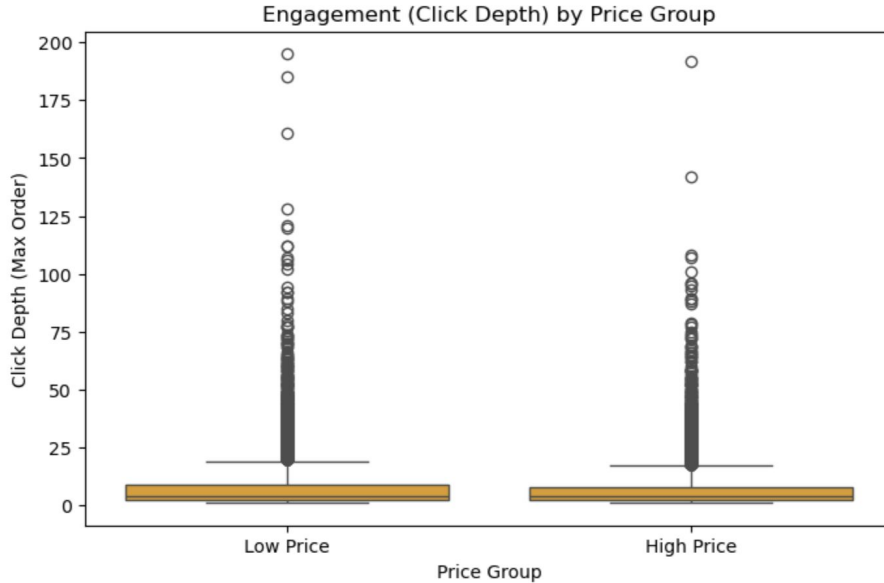
Analysis & Insights



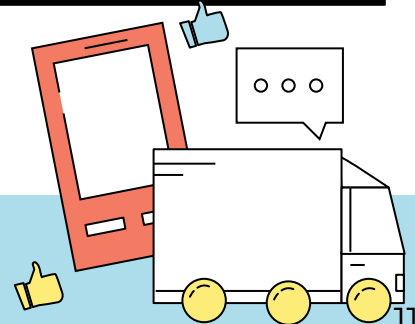


Step 1: Data Setup & EDA

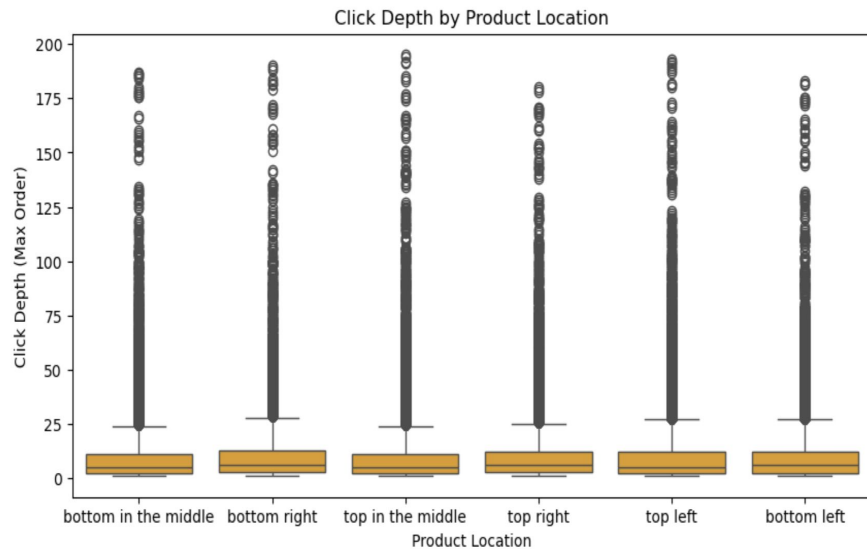
Product Price Influences Engagement



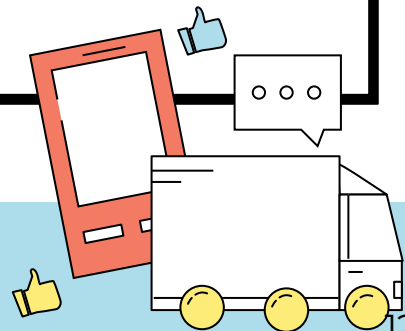
- Sessions with **lower-priced products show deeper click behavior** (avg. 7.0 clicks (low price) vs 6.7 clicks (high price))
- The difference is **small but statistically significant**



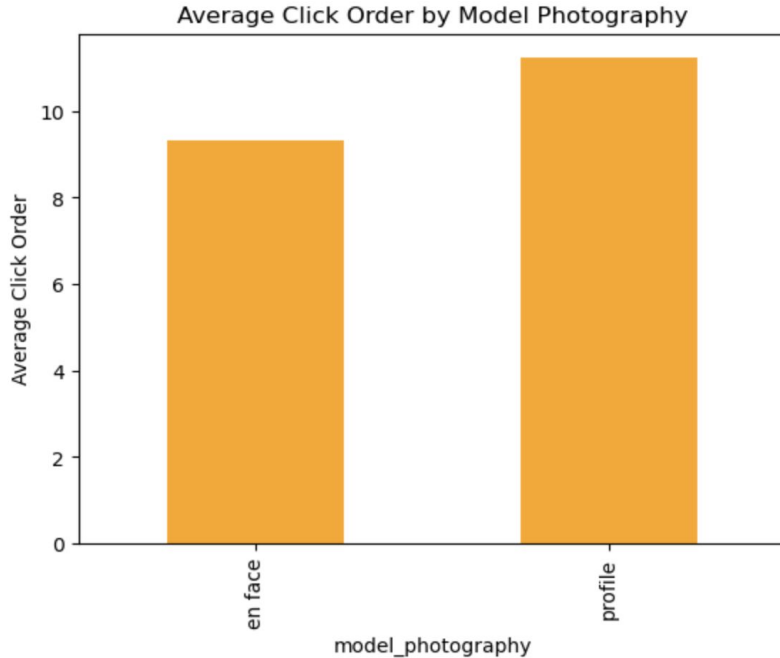
Product Presentation Affects Engagement



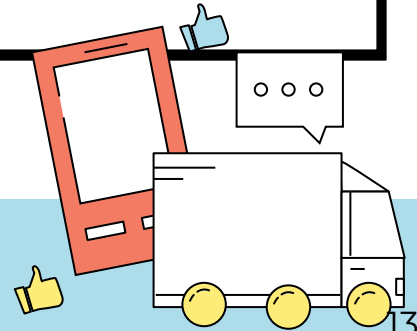
- Top middle and top left (195 clicks) **show slightly higher engagement**, suggesting that visual placement is important in online shopping



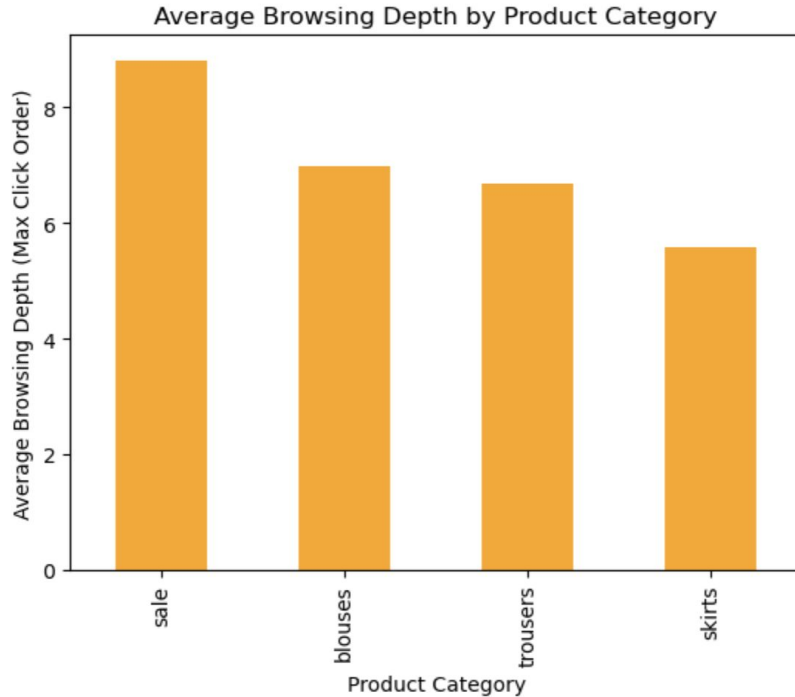
Model Photography Influences Early Clicks



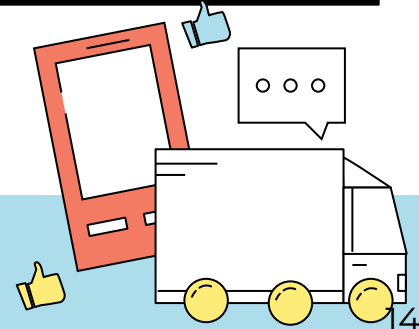
- En face images have a lower mean click order, meaning **en face images attracting clicks earlier** in the session
- Model photography style influences initial user engagement



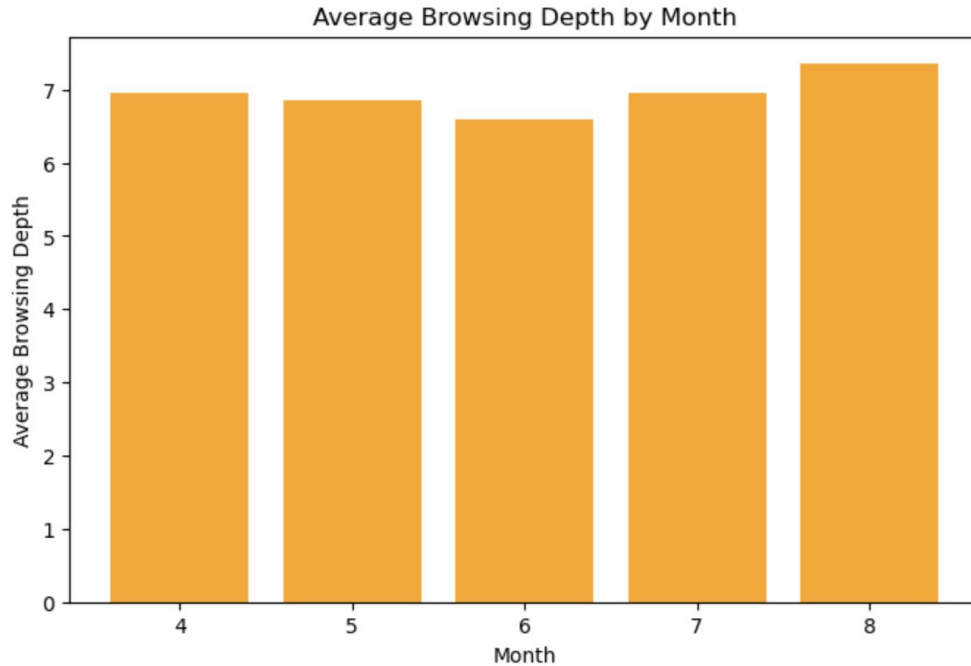
Category Drives Browsing Depth



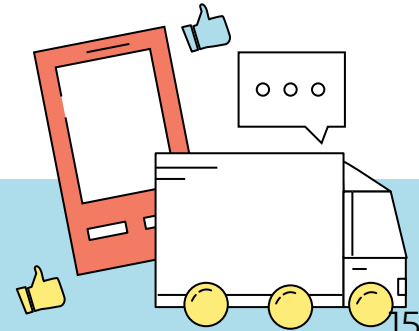
- Sale items drive the deepest engagement (avg. 8.8 clicks)
- Skirts show shallow browsing (avg. 5.6 clicks)



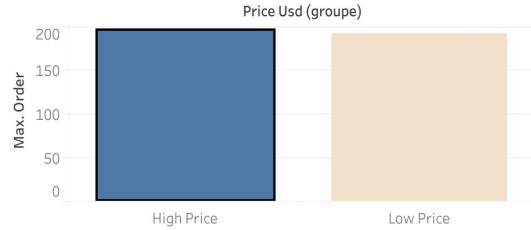
Seasonality Affects Browsing Depth



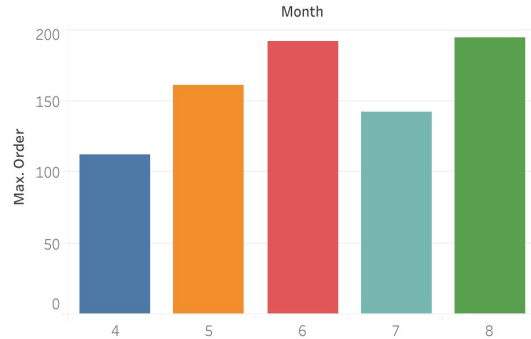
- Engagement is stable during most of the year
- Highest browsing depth in August



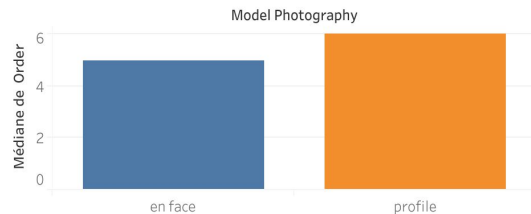
Product Price Influences Engagement



Seasonality Affects Browsing Depth



Model Photography Influences Early Clicks



Price Usd (groupe)
■ High Price
■ Low Price

Product Presentation Affects Engagement



Category Drives Browsing Depth

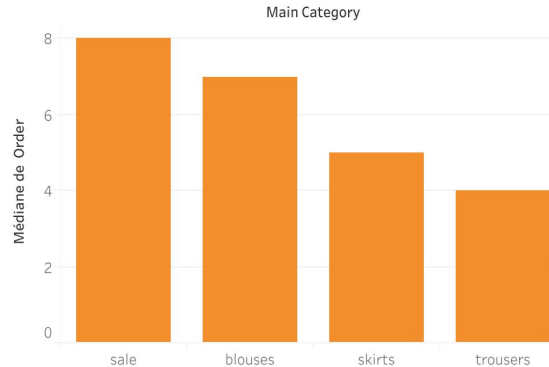


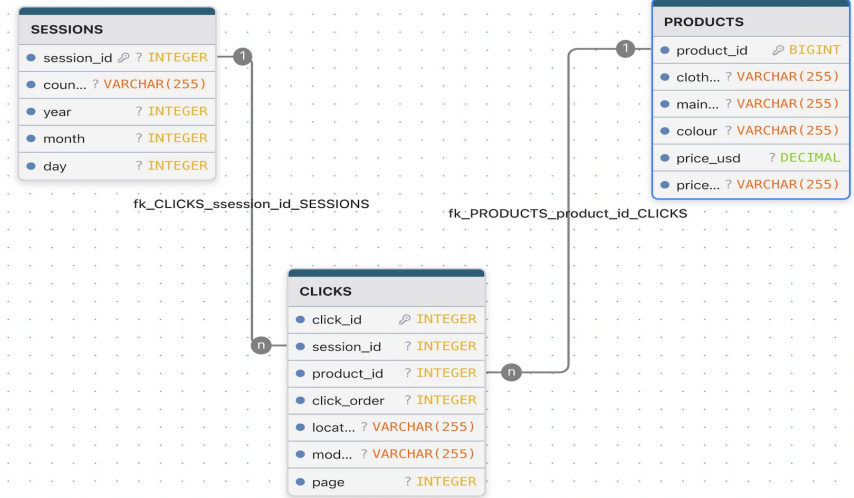
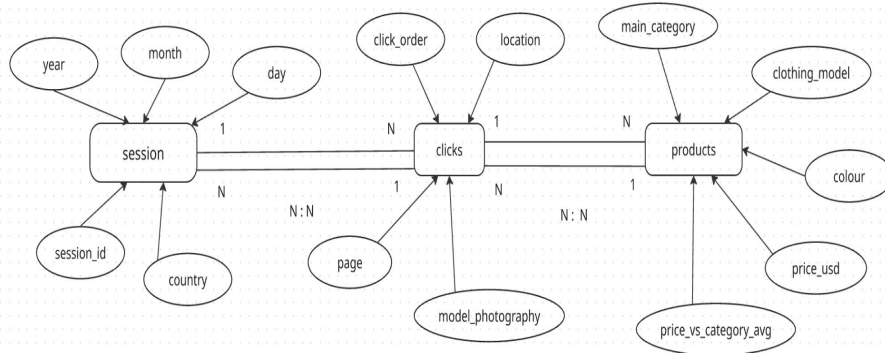
Tableau Visualizations

Tableau Visualizations

Step 2: Database



Database Creation & ERD



Administration

Schemas

Query 1

ecommerce_clickstream

SCHEMAS

Filter objects

ecommerce_clickstre...

Tables

Views

Stored Procedures

Functions

Info

Tables

Columns

Indexes

Triggers

Views

Stored Procedures

Functions

Grants

Events

Name	Engine	Version	Row Format	Rows	Avg Row Length	Data Length	Max Data Length	Index Length	Data Free
clicks	InnoDB	10	Dynamic	164826	73	11.5 MiB	0.0 bytes	12.5 MiB	4.0 MiB
products	InnoDB	10	Dynamic	217	75	16.0 KiB	0.0 bytes	48.0 KiB	0.0 bytes
sessions	InnoDB	10	Dynamic	23731	66	1.5 MiB	0.0 bytes	2.0 MiB	4.0 MiB

SQL Queries

```
2  -- Top 10 most popular products by total clicks
3  •  SELECT
4      product_id,
5      clothing_model,
6      main_category,
7      colour,
8      price_usd,
9      total_clicks,
10     unique_sessions
11  FROM product_popularity
12  ORDER BY total_clicks DESC
13  LIMIT 10;
```

100%

48:2

Result Grid

Filter Rows:

Export:

Fetch rows:

	product_id	clothing_model	main_category	colour	price_usd	total_clicks	unique_sessions
	3	B4	skirts	olive	52.00	3579	3212
☐	54	A2	trousers	blue	43.00	3013	2557
☐	30	A11	trousers	blue	62.00	2789	2454
☐	15	P1	sale	blue	38.00	2681	2266
☐	87	B10	skirts	black	67.00	2566	2415
☐	84	A4	trousers	blue	38.00	2522	2224
☐	91	A15	trousers	white	33.00	2489	2227
☐	55	A5	trousers	blue	43.00	2354	1949
☐	14	A10	trousers	blue	38.00	2280	1808
☐	47	A1	trousers	navy blue	28.00	2265	2104

```
31  -- Price analysis by category
32  •  SELECT
33      main_category,
34      COUNT(*) as product_count,
35      MIN(price_usd) as min_price,
36      AVG(price_usd) as avg_price,
37      MAX(price_usd) as max_price
38  FROM products
39  GROUP BY main_category
40  ORDER BY avg_price DESC;
```

100%

32:37

Result Grid

Filter Rows:

Export:

	main_category	total_products	total_clicks	avg_clicks_per_product	total_sessions
	trousers	43	49743	1156.8140	43305
	sale	82	38747	472.5244	34996
	blouses	59	38577	653.8475	35122
	skirts	33	38407	1163.8485	34922

final-project-486916 / ... / eshop_clothing_2008_processed / Tables / my_table

☆ my_table Requête Ouvrir dans

< Schéma Détails Aperçu Explorateur de tables Preview >

☐ Nom du champ	Type	Mode	Description
☐ year	INTEGER	NULLABLE	-
☐ month	INTEGER	NULLABLE	-
☐ day	INTEGER	NULLABLE	-
☐ order	INTEGER	NULLABLE	-
☐ country	STRING	NULLABLE	-
☐ session_id	INTEGER	NULLABLE	-
☐ main_category	STRING	NULLABLE	-
☐ clothing_model	STRING	NULLABLE	-
☐ colour	STRING	NULLABLE	-
☐ location	STRING	NULLABLE	-
☐ model_photography	STRING	NULLABLE	-
☐ price_usd	INTEGER	NULLABLE	-
☐ price_vs_category_avg	STRING	NULLABLE	-
☐ page	INTEGER	NULLABLE	-

Google BigQuery

```

1 SELECT
2   country,
3   COUNT(DISTINCT session_id) AS total_sessions,
4   AVG(`order`) AS avg_click_order
5 FROM
6   `final-project-486916.eshop_clothing_2008_processed.my_table`
7 GROUP BY country
8 HAVING total_sessions > 100
9 ORDER BY avg_click_order DESC;
```

✓ Requête terminée

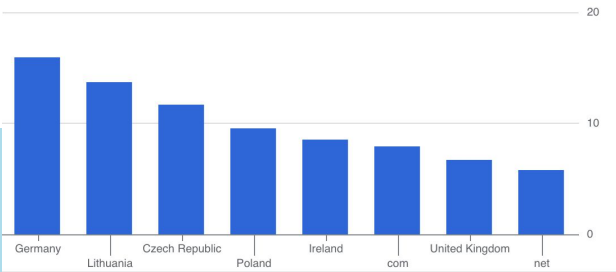
Utilisation du quota de traitement à la demande

Résultats de l...

+ Créer une conversation

Enregistrer

avg_click_order par country



< Informations sur le job Résultats Visualisation JSON

Ligne	country	total_sessions	avg_click_order
1	Germany	101	15.88968824940...
2	Lithuania	527	13.71009533121...
3	Czech Republic	2261	11.64383713825...
4	Poland	19582	9.502832871763...
5	Ireland	102	8.565967940813...

APIs

```
← → ↺ 🏠 ⓘ http://localhost:5000/api/sessions
Pretty print ✓
{
  "data": [
    {
      "country": "Poland",
      "day": 1,
      "first_click": 1,
      "last_click": 9,
      "month": 4,
      "session_id": 1,
      "total_clicks": 9,
      "unique_products": 9,
      "year": 2008
    },
    {
      "country": "Poland",
      "day": 1,
      "first_click": 1,
      "last_click": 10,
      "month": 4,
      "session_id": 2,
      "total_clicks": 10,
      "unique_products": 8,
      "year": 2008
    },
    {
      "country": "Ireland",
      "day": 1,
      "first_click": 1,
      "last_click": 6,
      "month": 4,
      "session_id": 3,
      "total_clicks": 6,
      "unique_products": 6,
      "year": 2008
    },
    {
      "country": "Ireland",
      "day": 1,
      "first_click": 1,
      "last_click": 4,
      "month": 4,
      "session_id": 4,
      "total_clicks": 4,
      "unique_products": 4,
      "year": 2008
    }
  ]
}
```

API	Description
GET api/sessions	Get all sessions
GET api/sessions/<id>	Get specific session details
GET /api/sessions/<id>/clicks	Get all clicks for a session
GET /api/products	Get all products
GET /api/products/<id>	Get specific product details
GET /api/products/popular	Get most popular products
GET /api/clicks	Get all clicks
GET /api/stats	Get database statistics

Step 3: Data Engineering



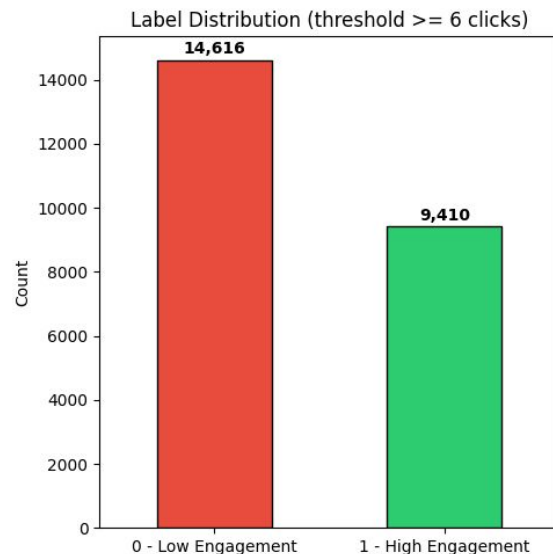
Feature Engineering

Feature Engineering

- Convert the click-based dataset into a **session-based dataset** by grouping events using the session ID
- **Create additional features** to better represent user behavior within a session (e.g., 'dominant_category', 'dominant_colour',...)
- Transform features:
 - Categorical features using One-Hot Encoding
 - Numerical features using Standard Scaling

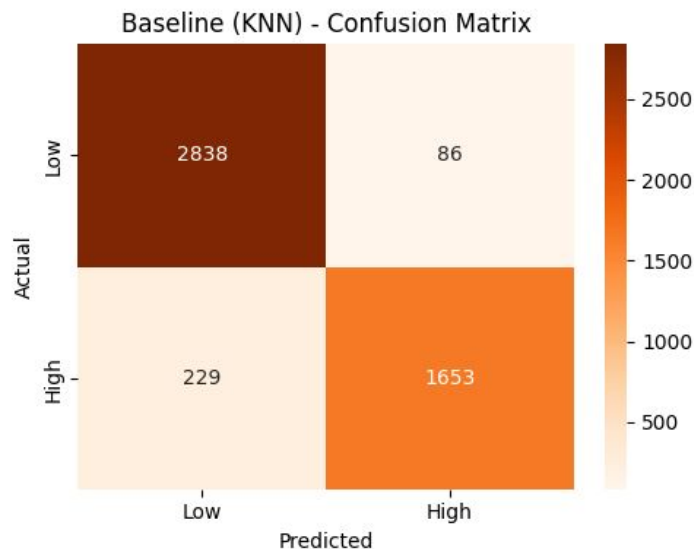
Label Creation

- Create a **binary engagement label** based on the total number of clicks per session
 - **Label 0** (low engagement): sessions with few clicks
 - **Label 1** (high engagement): sessions with many clicks (top 40%)
- Define engagement threshold (6 clicks) using click distribution

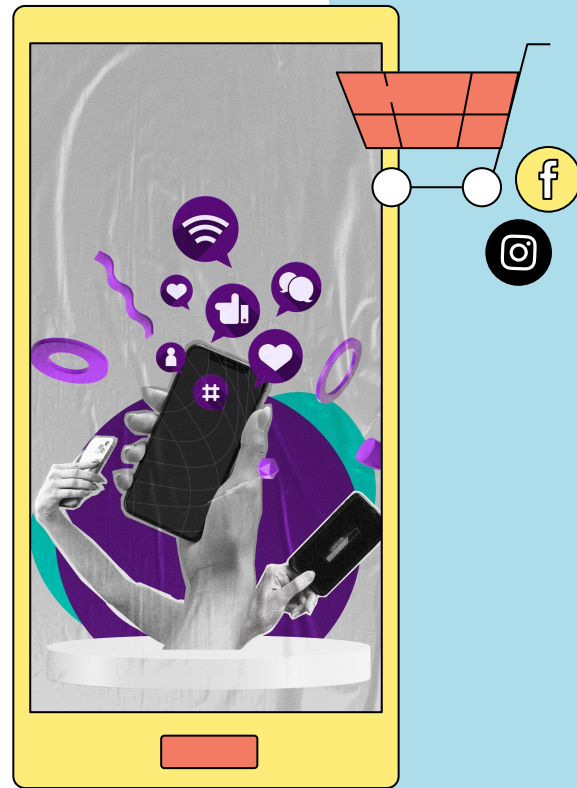
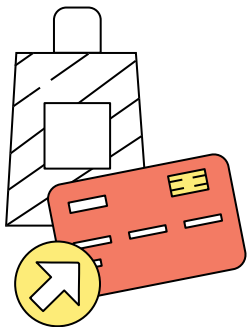


Baseline Model: KNN

- **Data split:** 80% training / 20% testing
- **Baseline model:** k-Nearest Neighbors (KNN)
- **Performance:** accuracy 93%
- KNN provides a **strong baseline**, especially for identifying low-engagement sessions, with room to improve recall on high engagement.

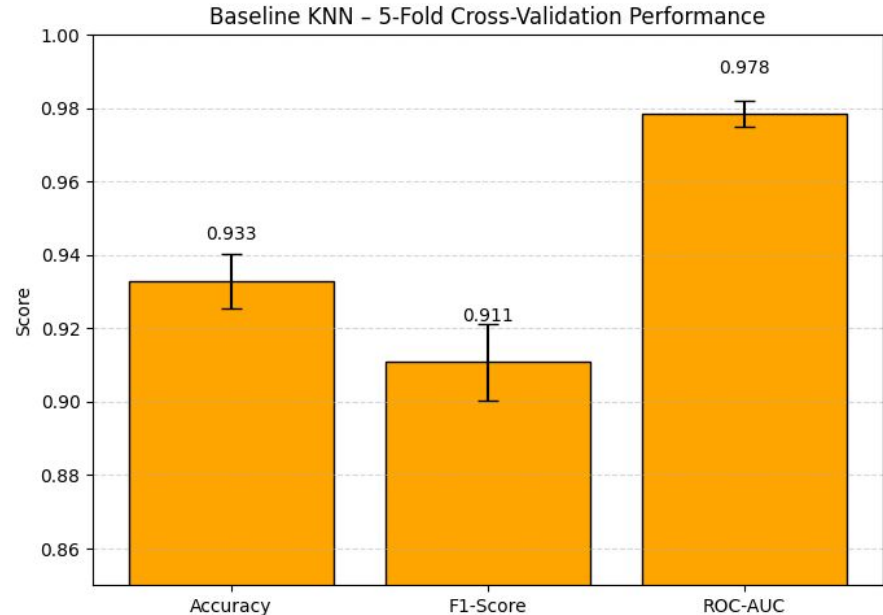


Step 4: Model Tuning

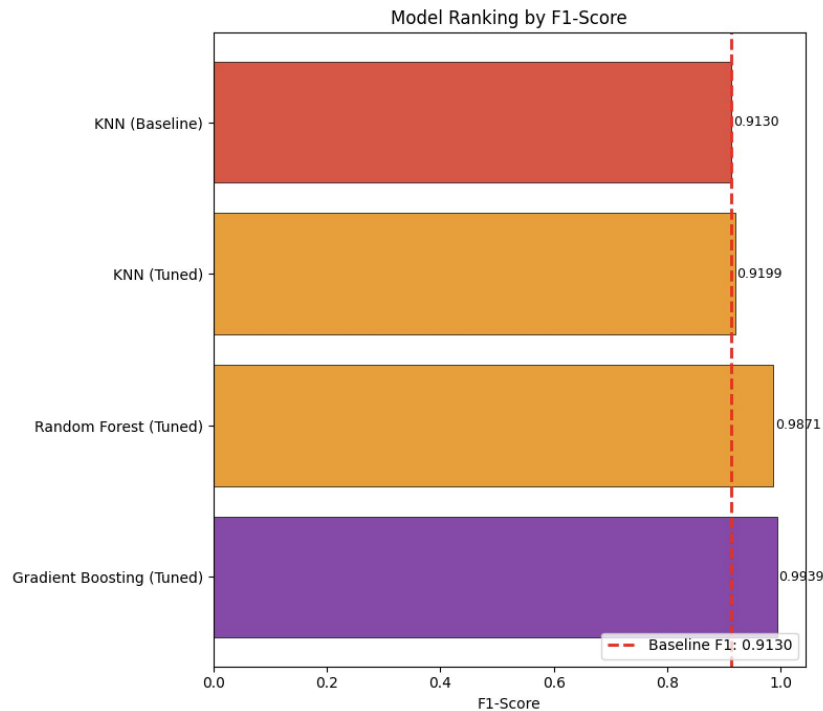


Cross-Validation

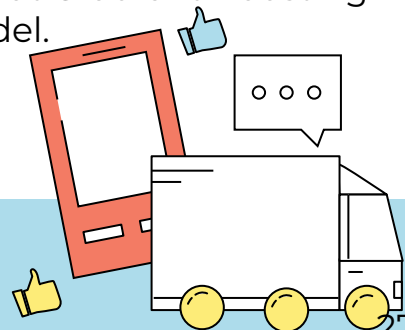
- **Method:** 5-fold cross-validation
- **Model:** KNN (baseline)
- **Results**
 - Achieves 93.3% accuracy and F1-score of 91.1%, indicating balanced performance across classes
 - The model is stable and generalizes well to unseen data
- Confirms KNN as a **reliable** and **solid baseline** for further model comparison



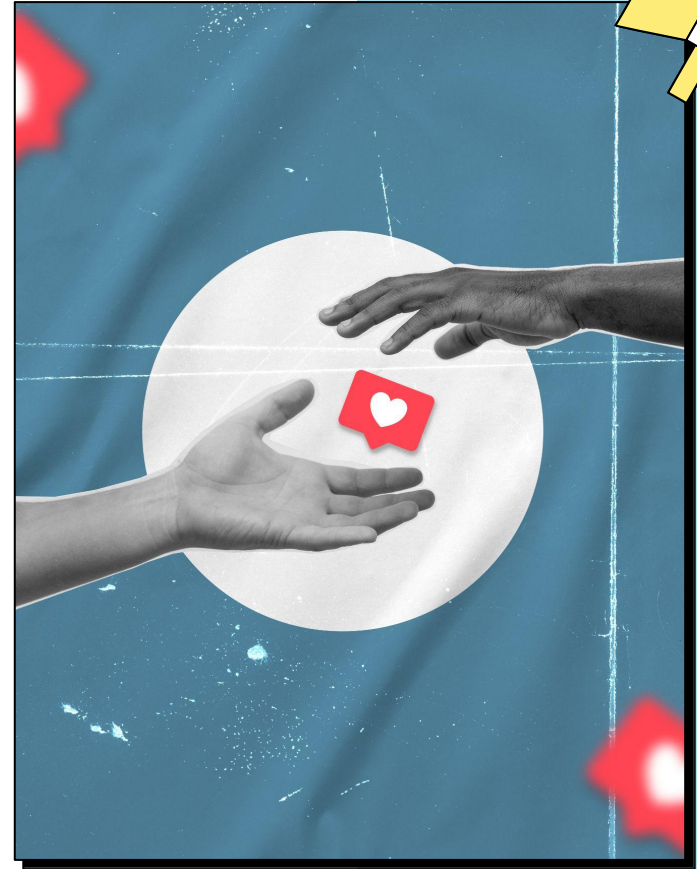
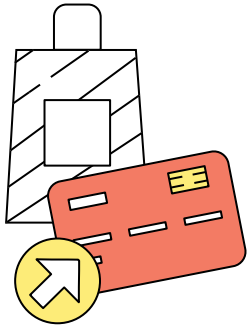
Hyperparameters Tuning



- **Method:** 5-fold Stratified Cross-Validation using GridSearchCV
- **Models tuned:** 3 classification models
- **Results**
 - Optimal hyperparameters identified for each model
 - **Best-performing model: Fine-tuned Gradient Boosting**
 - Achieved **99.4% accuracy**, significantly outperforming the baseline
- **Hyperparameter tuning greatly improves performance** and shows that Gradient Boosting is the best-performing model.



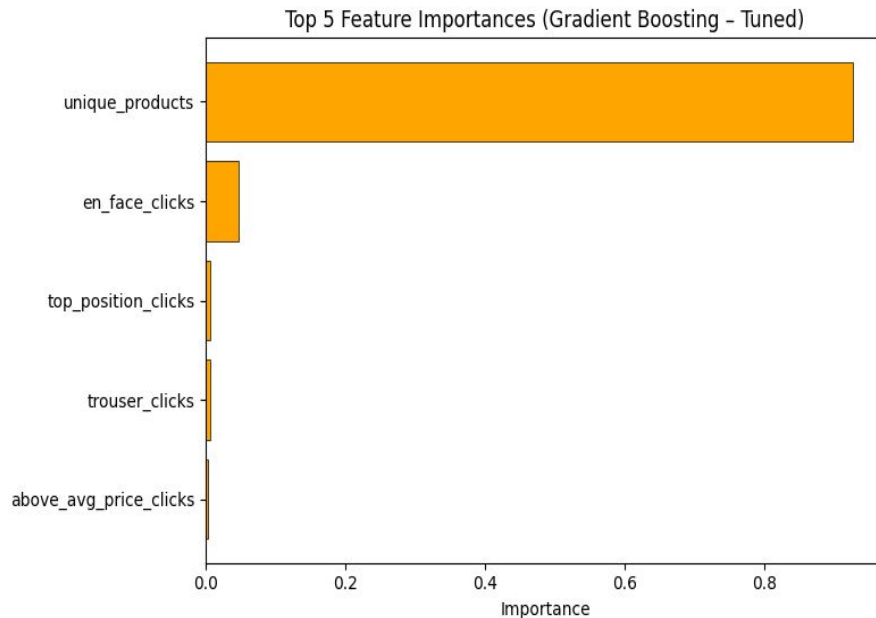
Step 5: Feature Impact



Feature Importance

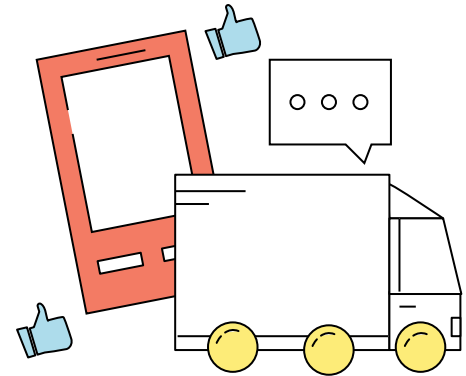
Extract feature importance from the best fine-tuned Gradient Boosting model, based on how much each feature reduces prediction error across all trees

- *'unique_products'* (~93%) is the **most influential feature**, indicating that the diversity of products a user interacts with strongly drives predictions
- *'en_face_clicks'* (~4.7%), capturing **user engagement with face-category products**
- Other features (<1%), such as *position_clicks* or category-specific clicks have minimal individual impact, suggesting the model relies more on overall product diversity than fine-grained click patterns



04

Conclusion

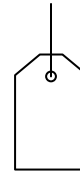
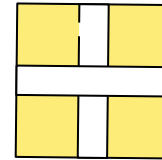
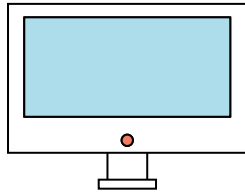
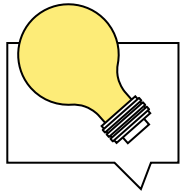


Conclusion

- **User engagement in online shopping can be predicted** effectively using machine learning
- **Price, product position, image style, category, and seasonality** influence browsing behavior
- **Product diversity viewed is the strongest factor** for high engagement
- Insights can help improve:
 - Product recommendations
 - Page layout and design
 - Marketing strategies
- **Data-driven decisions can improve the online shopping experience**

Resources

- Github Final Project
- Clickstream Data for Online Shopping Dataset
- Trello Dashboard
- Tableau Visualizations
- Database ERD
- Slide Templates



Thanks!

